

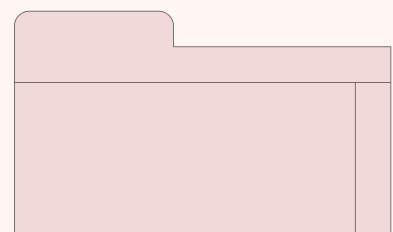
Project Name: Ellmuna App

Course: CS 411 – Software Engineering

Team: CS Year 4 – Group G1

Supervisor: Dr. Noorah Albehaijan

Date: December 2025



Team members:

Remas Alshehri – 2220005349

Joud Alotaibi – 2230006186

Jori Alghamdi – 2230002748

Jood Alali – 2230004483

Raya Al-Harbi – 2230005278

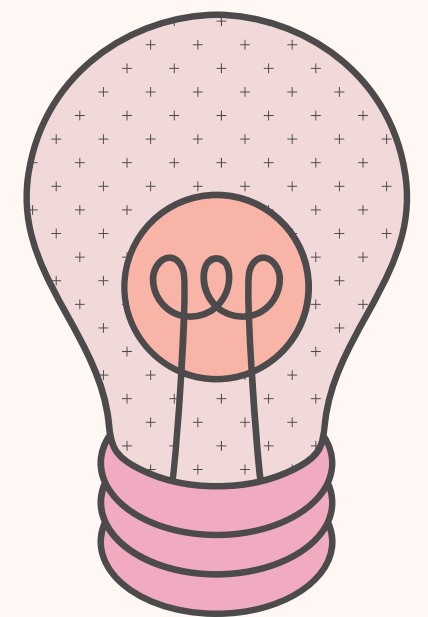
Raghad Alasfoor – 2230003688

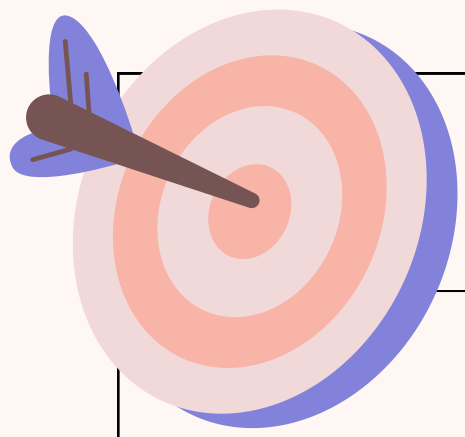


PROBLEM STATEMENT

Students in Saudi Arabia face difficulties when calculating their academic results or searching for universities and majors. Many available websites and applications use different calculation methods, which leads to confusion, stress, and sometimes inaccurate academic decisions.

In addition, academic information, especially for postgraduate programs, is scattered across multiple sources, which makes it difficult for students to find reliable and well-organized data in one place.





PROJECT GOAL

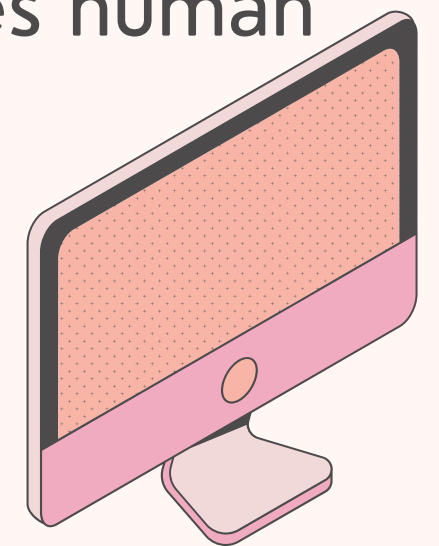
The goal of the Ellmuna App is to provide a unified and reliable academic platform that supports students across different educational levels in Saudi Arabia by addressing their academic calculation and information needs in one place.

Ellmuna aims to help high school and university students accurately calculate their academic results using standardized and trusted calculation methods. In addition, the application enables students to explore Saudi universities and available majors for Bachelor's, Master's, and PhD programs through a well-organized and user-friendly interface. By centralizing these services, Ellmuna reduces confusion, saves time, and supports students in making informed academic decisions.

PROJECT VALUE AND IMPORTANCE

This project matters because Saudi students currently rely on multiple scattered platforms to perform essential academic tasks, such as calculating GPA, weighted admission scores, and searching for suitable universities and programs. These fragmented resources often lead to confusion, inaccurate calculations, and wasted time.

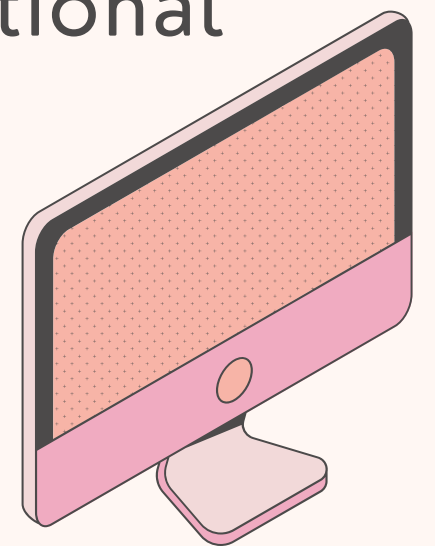
Ellmuna addresses this issue by providing a single, unified platform that centralizes academic calculations and reliable university information. By automating GPA and weighted score calculations using standardized formulas, the system reduces human error and helps students better understand their academic standing.



PROJECT VALUE AND IMPORTANCE

Additionally, Ellmuna helps students at different educational levels easily understand and compare their academic options within one system. This makes academic planning simpler, saves time, and reduces stress by providing clear and accurate academic services.

Furthermore, Ellmuna aligns with Saudi Arabia's Vision 2030 by supporting digital transformation in education. The project uses technology to improve access to academic support and enhance the overall quality of educational services for students.



HOW WE DID IT (APPROACH/METHODOLOGY)

- Process model: We followed an Agile iterative approach (sprints) to deliver features incrementally, then review and test them.
- Documentation-driven workflow: We started with SRS (requirements), then SDS (design/architecture), then STS (testing plan & criteria).
- Implementation scope: Four core modules: Weighted Score, GPA (4.0 & 5.0), Universities/Programs browsing, and Settings (Dark mode / Welcome screen).
- Tools: We used Figma to design the UI screens and prototype the main flows, and we relied on the tools listed in our SPMP for implementation, documentation, and testing.

HOW WE DID IT (APPROACH/METHODOLOGY)

- Project coordination: We used a high-level timeline to track submission dates and milestones, and managed day-to-day work via our WhatsApp group (task ownership per deliverable, file sharing, continuous updates).

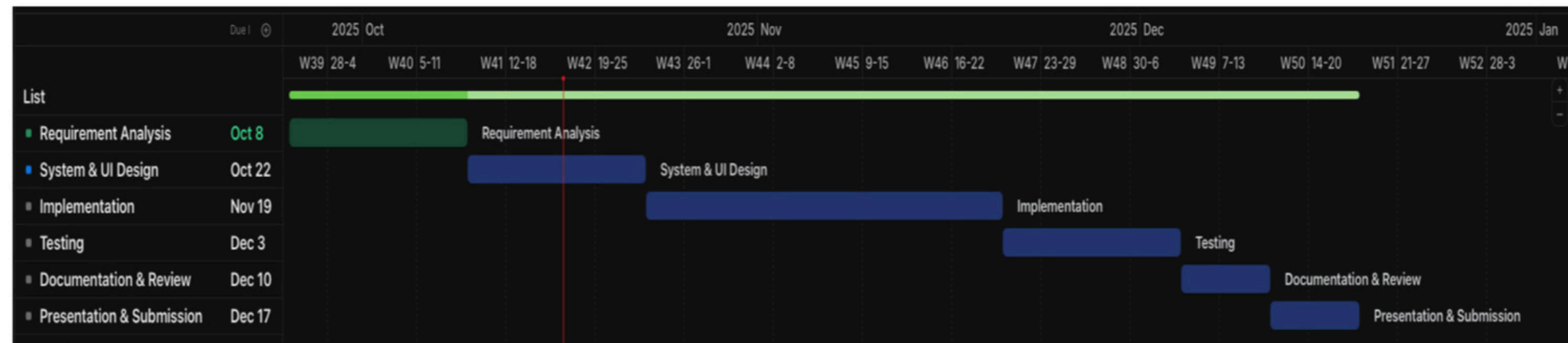


Figure 6 Ellmuna Project Schedule

Part 1 - Joud S ✓
Part2 -Joud S ✓
Part3 - jori ✓
Part 4 - Raya ✓
Part5 - Raghad ✓
Part6 - Remas ✓
Part8 - Jood alali ✓
Part 7 - Jori ✓
Part9 - Remas ✓
Part 10 - Raghad ✓

تم التحرير 5:05 م

QUICK LOOK AT THE KEY DOCUMENTS

SPMP (Software Project Management Plan)

- Explains how the project was managed: Agile process model, team roles, coordination, risks, and deliverables.
- Includes: Deliverables & schedule allocation + risk/team roles.

No.	Name	Main Responsibilities
1	Remas Alshehri	Project Coordination and Supervision: This member coordinates project activities, allocates tasks, plans meetings, reads documents, and coordinates with the supervisor.
2	Raya Al-Harbi	Documentation & Support: This member assists in documentation and report preparation, works on design and test tasks, and assists in preparing the final presentation.
3	Raghad Al-Asfoor	System Analysis & Core Implementation: This role is critical. It involves running the system analysis, applying the GPA/CGPA calculation logic, and then testing the accuracy of those results very carefully.
4	Joud Al-Otaibi	UI/UX Design & Visuals: This person is responsible for the actual UI/UX design of the user interface and overall user experience, including helping with the application's general visual coordination.
5	Jori Al-Ghamdi	Technical Document Preparation: This role is all about the paperwork. It includes preparing and constantly updating all technical documents (Idea, Proposal, SPMP, Final Report) and taking an active part in every team review.
6	Jood Al-Ali	Testing & Presentation: This member takes the lead on conducting all system testing, documenting the results, and assists the entire team with preparing the final presentation.

13 Risk Management Activities

Activity	Frequency	Responsible Party
Initial Risk Identification	At project initiation	Project Manager & Team
Risk Assessment	Bi-weekly	Project Manager
Risk Register Update	Weekly	All Team Members
Risk Review Meeting	Every two weeks	Entire Team
Final Risk Review	At project closure	Project Manager

QUICK LOOK AT THE KEY DOCUMENTS

SRS (Software Requirements Specification)

- Documents functional + non-functional requirements, user classes, and main system features.
- Includes main use cases (Weighted Score, GPA, Universities browsing, Settings).
- Defines performance targets such as 2 seconds for calculations and 3 seconds for university/program queries.
- Covers security,privacy expectations and input validation, with PDPL considerations.

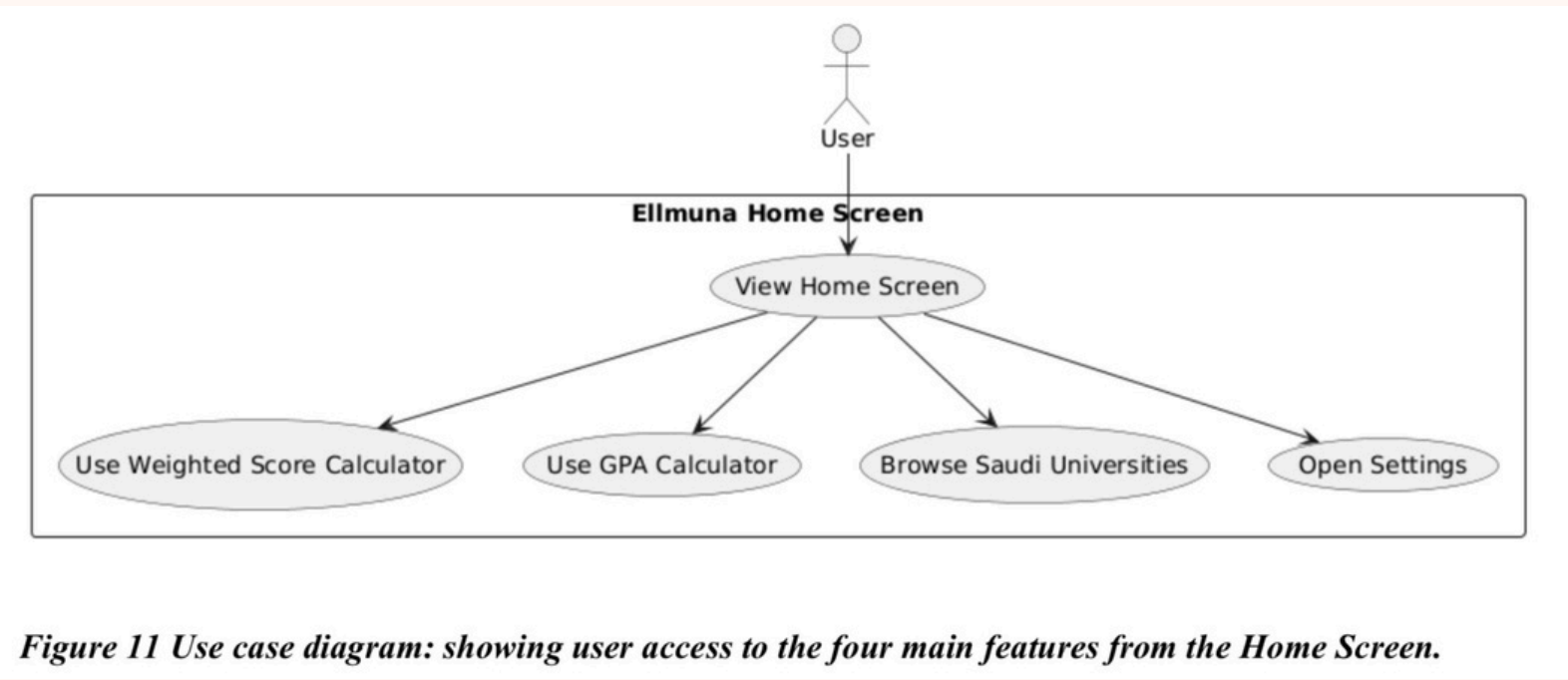


Figure 11 Use case diagram: showing user access to the four main features from the Home Screen.

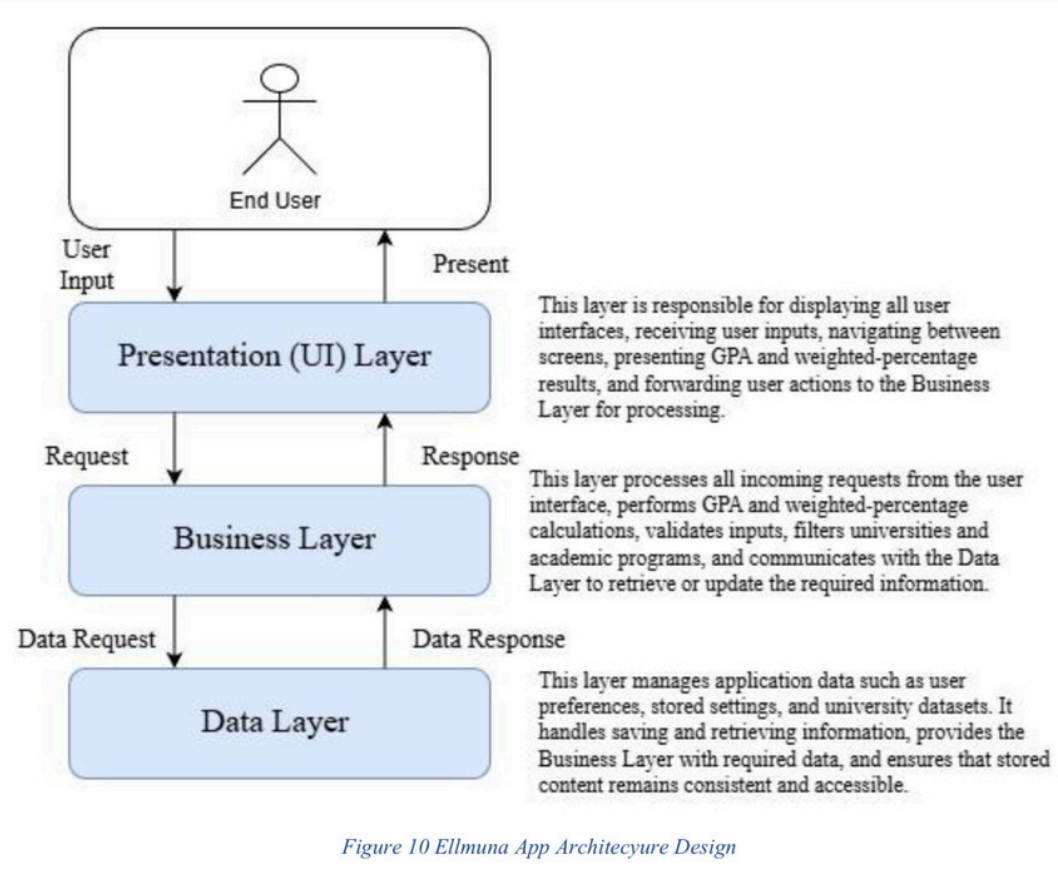
User Class	Description / Characteristics
High School Students	Primary users use the app to calculate their weighted score and view university eligibility. Have basic mobile skills.
University Students (Bachelor)	Use the GPA calculator to track semester and cumulative GPA. Moderate computer literacy.
Postgraduate Students (Master's/PhD)	Use the app mainly to browse and compare programs offered by Saudi universities.
General Users / Parents	May browse university data for guidance; use the app occasionally.
Admin (Future Role)	May update data or maintain database content in future versions.

Table 1 User Classes and Characteristics

QUICK LOOK AT THE KEY DOCUMENTS

SDS (Software Design Specification)

- Transforms SRS requirements into a structured design blueprint (architecture, components, interfaces, data structures).
- Implements a clear browsing hierarchy: Degree Level → University → College → Program.
- Includes a Requirements Traceability Matrix mapping SRS requirement IDs to design components.
- Documents UI screen flows for major features.



Association ID in SRS	Technical Assumptions / Customer Needs	Functional Requirement	User	Association ID in SDS	System Component
REQ-1	Valid device, app installed	Display Home Screen	General User	7.1	Homepage UI
REQ-2	Stable UI rendering	Show logo, name, buttons	General User	7.1	Homepage UI
REQ-3	Navigation enabled	Correct redirection	General User	7.1	Homepage UI
REQ-4	Usable interface	Simple user-friendly UI	General User	7.1	Homepage UI
REQ-5	Valid HS inputs	Input three academic scores	High School Student	7.3	Weighted Calculator UI
REQ-6	Default values loaded	Use default weights	High School Student	7.3	Weighted Calculator UI
REQ-7	Correct formula	Calculate weighted score	High School Student	7.3	Weighted Calculator Logic
REQ-8	Result evaluation	Display evaluation message	High School Student	7.3	Weighted Result UI
REQ-9	Input validation	Handle invalid input	High School Student	7.3	Error Handling Module
REQ-10	Select scale available	Select GPA scale	University Student	7.4	GPA Scale UI
REQ-11	Correct GPA formula	Calculate GPA accurately	University Student	7.4	GPA Logic
REQ-12	Clear results	Display GPA results	University Student	7.4	GPA Result UI
REQ-13	Validation	Handle invalid course data	University Student	7.4	Error Handling Module
REQ-14	Temp session	Maintain recalculation	University	7.4	Session

QUICK LOOK AT THE KEY DOCUMENTS

STS (Software Testing Specification / Test Plan)

- Provides the overall test plan and levels (component, integration, interface, security, performance, recovery, regression, acceptance, beta).
- Includes testing process elements such as test deliverables, tasks, responsibilities, schedule, and pass/fail criteria.

5.1 Component Testing

Item	Details
Inputs	• Valid, invalid, and boundary exam score values • Different GPA and credit-hour combinations • Sample university and college data
Outputs	• Accurate weighted score calculations • Correct GPA outputs • Proper validation and error messages • Correct handling of invalid or unexpected inputs
Success Criteria	• All outputs match expected formulas • No logical or arithmetic inconsistencies • Input validation prevents invalid data from progressing
Techniques	• Unit testing • Boundary value analysis • Input validation testing
Completion Criteria	• All unit tests pass • Zero unresolved logic errors • All functions meet their defined behavior

Table 6 Component Testing Summary

Task	Date
Review SRS	Nov. 22, 2025
Complete SDS	Nov. 24, 2025
Develop test cases	Nov. 26, 2025
Execute testing activities	Nov. 28, 2025
Troubleshoot errors	Nov. 30, 2025
Modify the system	Dec. 2, 2025
Prepare STP	Dec. 4, 2025

Table 18 schedule of the testing tasks.

6. PASS / FAIL CRITERIA

In this section, the pass/fail criteria that will be used in the Ellmuna system are described. These criteria will help decrease the chances of failure and increase the efficiency of the system. The system must process all GPA and weighted score calculations correctly, display the correct university information, and ensure easy, smooth navigation between all sections. Only then can all the test cases pass without functional errors to allow the running of the application reliably with user-friendly experience. Just in case of any runtime issue, it must be handled by the system without faults. This must be achieved to ensure that the system operates efficiently and effectively, giving accurate academic results with Ellmuna.

6.1 Suspension Criteria

Suspension criteria are the conditions under which testing cannot continue either forward or backward, pausing a part of or the whole testing activity. Even if a portion is suspended, other sections unrelated to that may continue testing smoothly. Herein, conditions that might lead to the suspension of the testing process in Ellmuna could be:

- When a critical failure occurs, which prevents GPA or weighted score calculations from running.
- Lack of availability of the university dataset or failure retrieving it during execution.
- When identical inputs yield different results in different test attempts:
- Failure to connect with the cloud database that houses university programs.
- System crashes during navigation between sections or when switching between calculation modes.
- Shutting down the development or test environments during scheduled maintenance or holidays.

TESTING STRATEGY & RESULTS

Unit Testing:

Each system component is tested individually to verify correct calculations, input validation, and expected outputs.

Integration Testing:

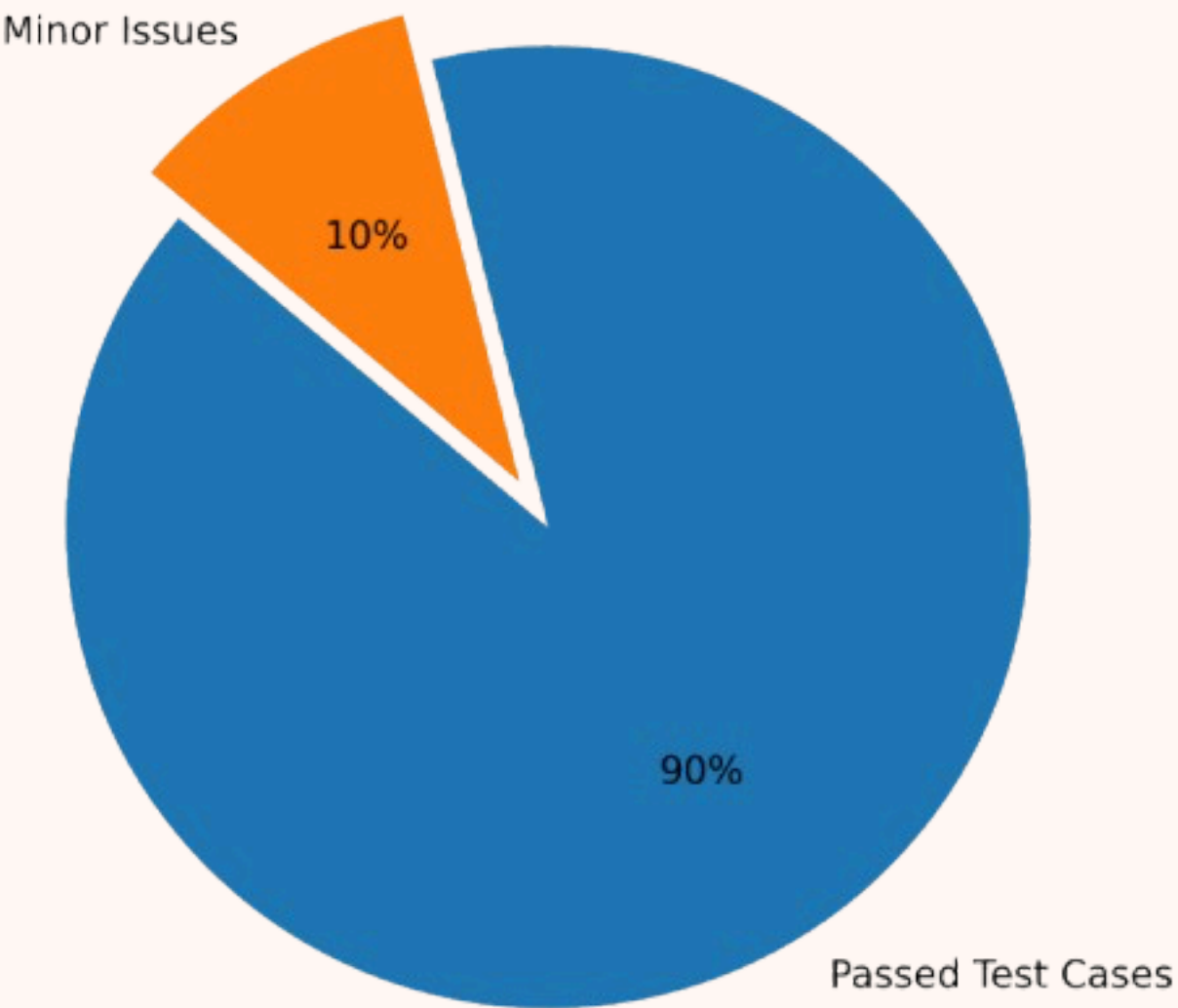
Testing the interaction between system modules to ensure smooth data flow, correct navigation, and consistent behavior across features.

Acceptance Testing:

Testing the system using realistic user scenarios to confirm that all requirements are met and the system is ready for final approval.

RESULTS

Overall Testing Results



CHALLENGES

- **Time and Team Coordination:** Several important documents with limited time.

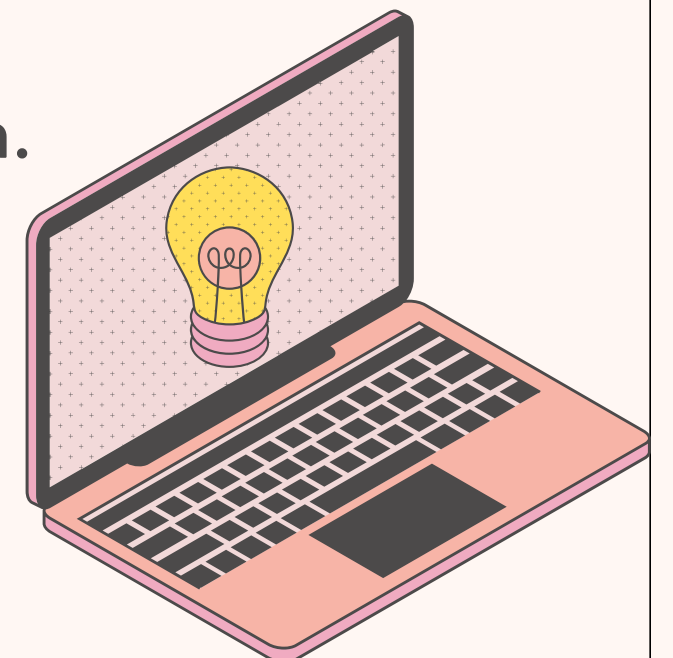
Solution: Task distribution and adherence to the project schedule defined in the SPMP.

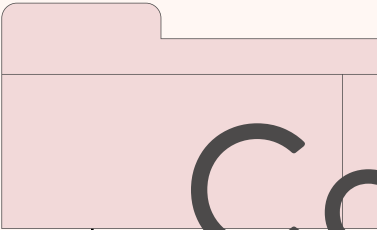
- **Lack of Standardization in Academic Systems:** Academic systems and calculation methods vary across universities, which makes it difficult to apply a single unified calculation approach.

Solution: The system was designed to support multiple academic calculation standards, allowing flexibility to accommodate different university grading systems.

- **Keeping Academic Information Up to Date:** Academic programs and requirements may change over time, which can affect data accuracy.

Solution: The system supports periodic updates to maintain accurate and reliable information.





Conclusions — Summary of Key Findings

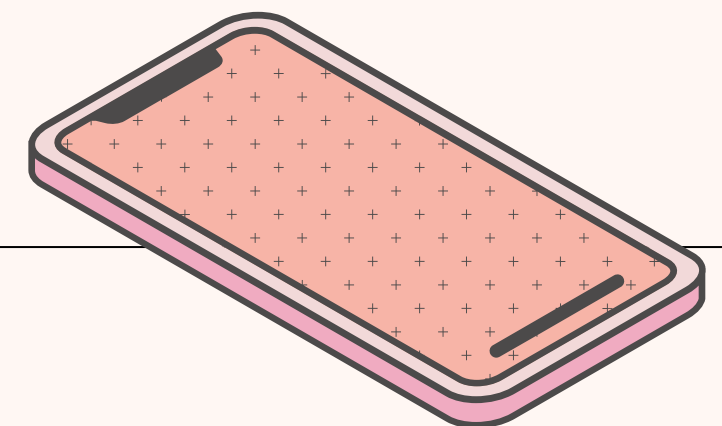
1. The Ellmuna project successfully delivered a unified academic application that serves students across multiple educational levels in Saudi Arabia.

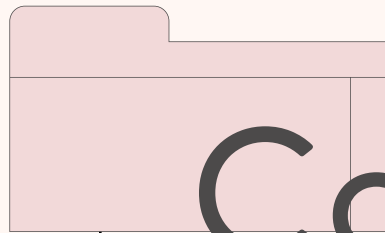
2. The system provides accurate GPA and weighted score calculations based on officially adopted formulas, reducing confusion caused by inconsistent external tools.

3. A well-structured architecture was designed following IEEE standards, ensuring clear separation between presentation, logic, and data layers.

4. Comprehensive documentation (SRS, SDS, SPMP, and STP) ensured that requirements, design, management, and testing were fully aligned.

5. Testing results confirmed that the core functionalities operate accurately, efficiently, and with acceptable performance and usability.





Conclusions — Future Work

1. Integrating the application with official Saudi university APIs to provide real-time admission requirements and updated program data.

2. Adding a user account system to store academic history and track progress over time.

3. Expanding features to include notifications for deadlines, admission periods, and academic alerts.



4. Enhancing analytics and reports to provide personalized academic recommendations.



5. Conducting broader user testing to further improve UI/UX and system scalability.

**THANK YOU FOR
YOUR ATTENTION**